

BRIAN BELL

Regulated Manufacturing ◇ System Design ◇ Robotics ◇ Automation
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EXPERIENCE

Custom Manufacturing & Engineering

Oct 2025 - Current

Mechanical Engineer

Tampa, FL

- Managing several concurrent manufacturing programs end-to-end, frequently inheriting distressed programs; responsible for contract breakdown, sourcing, execution, and customer reporting through production delivery.
- Fluent in authoring and interpreting engineering drawings, navigating requirement and test specifications, and coordinating with machine shops from prototype through production delivery. Achieved first article approval with zero redlines; sourced prototype chassis fabrication at \$300 against competing quotes of \$2,500 and \$9,000.
- Performed electromechanical system redesign and testing in light of obsolescence issues. Lowered the cost and complexity to manufacture staple systems, while improving our technical data in parallel.
- Conducted preliminary research and conceptual design across multiple robotics programs, including modular military payload rover systems spanning locomotion, sensors, and payload integration.
- Subject matter expert for reflow soldering profile development and thermal chamber testing across product lines; also leading reinstatement of vibration table testing capabilities.

MWA Environmental

June 2024 - December 2024

Cadet Engineer

Brisbane, QLD

- Built Python automation systems to parse and structure high-volume datasets (200+ weather stations; 43,200+ octave-band noise measurements), enabling filterable analysis and compliance-driven threshold comparison.

PROJECTS

Autonomous Cobot-Enabled Workcell

Oct 2025 - Present

Custom Manufacturing & Engineering

Tampa, FL

- Technical Lead for a commercial high-precision dual-cobot manufacturing platform supporting tele-operated and autonomous operation. Drove development across system architecture, electromechanical integration, control software, sensor fusion, OpenCV/ArUco visual servoing, and Isaac Sim digital twin. Interfaced directly with executive leadership and clientele. Contributed to R&D proposals for continued development.

STOA — Research Knowledge Graph

2026 - Present

Personal Project

- Building a full-stack academic knowledge graph platform as a research tool: arXiv and PDF ingestion, Claude API paper concept extraction, sentence-transformer embeddings for similarity scoring, and Neo4j graph storage. Designed pipeline, schema, and architecture with no prior Neo4j experience.

EDUCATION

Queensland University of Technology, Brisbane

June 2025

B.Eng. Honors Mechanical | Minors in Computer Science & Finance

GPA: 7.0 | First Class Honors | Executive Deans' List for Academic Excellence

TECHNICAL STRENGTHS

Compliance	AS9100, CMMC FedRAMP
Skills	Systems Engineering, Technical Drawing, GD&T, Failure Analysis, Electromechanical Assembly
Tools	SolidWorks, ANSYS, Universal Robots, ROS2, Isaac Sim, Linux, OpenCV, TensorFlow
Languages	Python, C++, SQL, Neo4j
Equipment	Reflow Soldering, Thermal Chamber, Vibration Testing, 3D Printing